Joint Distributions

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MATH 241

Week 11, Lecture #1 — Ch. 11.1

Final Stretch

We're almost there!! Here's a breakdown of what's left:

- Ch. 7: Joint distributions
  - Nov. 12th, 17th, 19th
  - CP6 on Thursday 11/19, HW8 due Monday 11/23
- Ch. 9: Conditional expectation
  - Nov. 24th
- Ch. 10: Inequalities (Law of Large Numbers; Cental Limit Theorem)
  - Dec. 1st, 3rd
  - HW9 due Friday 12/4

Final Stretch

- Final project
  - Progress checkpoint: due Friday 11/20
  - Demonstrations on Tuesday 12/8 in lieu of class
  - Final portfolio by Wednesday 12/9
  - Time to work in class: Thurs. 11/19, Thurs. 12/3 (tentatively)
- Final exam
  - Wednesday 12/16 from 1-3pm in Rocky 300

Single RVs

We've looked at situations involving just **one** single random variable. However, we might look at **two** random variables simultaneously and identify items such as the CDF, PDF, probabilities, and moments as before. We will look at:

- Discrete X and discrete Y
- Continuous X and continuous Y
- A combination of a discrete RV and continuous RV

Motivating Example #1

Suppose we were looking at the distribution of Dutchess County voters in the 2026 midterms. We may look at age group and whether someone identifies as white as two potential random variables.

$$f_X(x) = \begin{cases} 1 & \text{if under 30} \\ 0 & \text{otherwise} \end{cases}$$

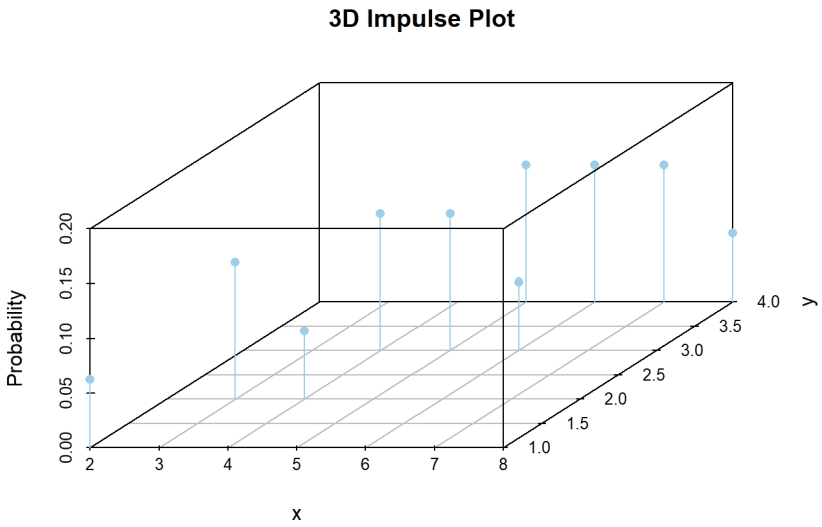
$$f_Y(y) = \begin{cases} 1 & \text{if white} \\ 0 & \text{otherwise} \end{cases}$$

Motivating Example #2

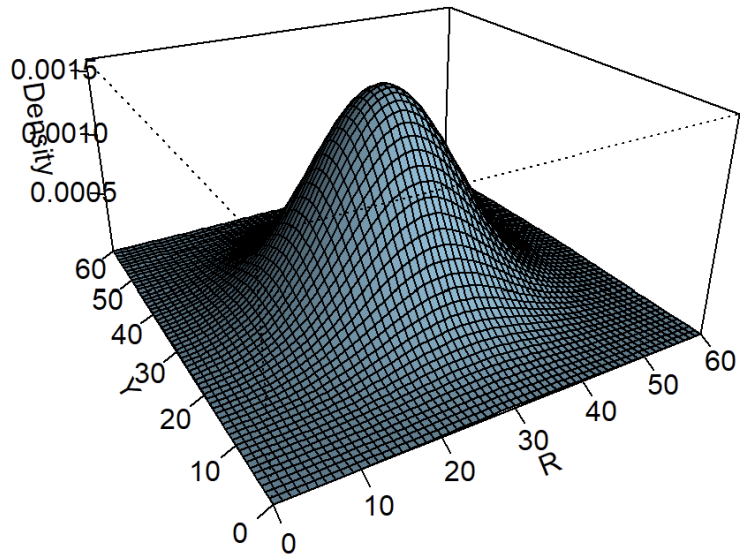
Suppose we were looking at the distribution of runners in a marathon. We can look at age and running times as two different random variables.

$$X \sim Normal(\mu = 32, \sigma = 11)$$
$$Y \sim Normal(\mu = 240, \sigma = 40)$$

Visuals



Discrete Joint RV



Continuous Joint RV

- (Ross, 2022)

Discrete Joint RVs - Joint CDF

Recall: For a single disjoint RV X, its CDF was:

$$F_X(x) = P(X \leq x)$$

Definition (7.1.1: Joint CDF)

The joint CDF of discrete RVs X and Y is the function $F_{X,Y}$, given by:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

- Will consist of jumps and flat regions

Discrete Joint RVs - Joint PMF

Recall: For a single disjoint RV X, its PMF was:

$$f_X(x) = P(XI = x)$$

Definition (7.1.2: Joint PMF)

The joint PMF of discrete RVs X and Y is the function $f_{X,Y}$, given by:

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

- Nonnegative (e.g. $P(X = x, Y = y) > 0$)
- Probabilities sum up to 1:

$$\sum_x \sum_y P(X = x, Y = y) = 1$$

Discrete Joint RVs - Marginal PMF

From a joint distribution, we may want to look back at just a single distribution. We call this single distribution a **marginal** distribution.

Definition (7.1.3: Marginal PMF)

For discrete RVs X and Y, the marginal PMF of X is:

$$P(X = x) = \sum_{y} P(X = x, Y = y)$$

- Synonymous with the PMF of X
  - Results from looking at X and Y together initially

Discrete Joint RVs - Conditional PMF

From a joint distribution, we may care about point probabilities of one random variable given the value of a different random variable. We call this single distribution a **conditional** distribution.

Definition (7.1.4: Conditional PMF)

For discrete RVs X and Y, the conditional PMF of Y given $X = x$ is:

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

- Another valid PMF of Y, but for a given value of x

Example: Free Throws Home and Away

Suppose we look at Mitchell Robinson's free throws from the 2025-26 season. We will let X denote a random variable indicating a made free throw or missed free throw, and Y denote a random variable indicating the location of the free throw (home vs. away).

The use of indicators for multiple random variables is characteristic of a **contingency table**, where $X_i = I(\text{some condition})$.

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In our case, we let:

- $X \sim I(\text{Mitchell makes a free throw})$
- $Y \sim I(\text{at home})$

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- $Y \sim I(\text{at home})$

	$X = 0$	$X = 1$	Total
$Y = 0$	$\frac{35}{101}$	$\frac{20}{101}$	$\frac{55}{101}$
$Y = 1$	$\frac{25}{101}$	$\frac{21}{101}$	$\frac{46}{101}$
Total	$\frac{60}{101}$	$\frac{41}{101}$	$\frac{101}{101}$

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$Y=0$	$\frac{35}{101}$	$\frac{20}{101}$	$\frac{55}{101}$
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Total	$\frac{60}{101}$	$\frac{41}{101}$	$\frac{101}{101}$

For the joint distribution of X, Y:

$$f_{X,Y}(0,0) = P(X=0, Y=0) = \frac{35}{101}$$

$$f_{X,Y}(0,1) = P(X=0, Y=1) = \frac{25}{101}$$

$$f_{X,Y}(1,0) = P(X=1, Y=0) = \frac{20}{101}$$

$$f_{X,Y}(1,1) = P(X=1, Y=1) = \frac{21}{101}$$

Example: Free Throws Home and Away

	$X = 0$	$X = 1$	Total
$Y = 0$	$\frac{35}{101}$	$\frac{20}{101}$	$\frac{55}{101}$
$Y = 1$	$\frac{25}{101}$	$\frac{21}{101}$	$\frac{46}{101}$
Total	$\frac{60}{101}$	$\frac{41}{101}$	$\frac{101}{101}$

For the marginal distribution of X:

$$f_X(0) = \sum_{y=0}^{1} P(X = 0, Y = y) = \frac{35}{101} + \frac{25}{101} = \frac{60}{101}$$

$$f_X(1) = \sum_{y=0}^{1} P(X = 1, Y = y) = \frac{20}{101} + \frac{21}{101} = \frac{41}{101}$$

Example: Free Throws Home and Away

	$X=0$	$X=1$	Total
$Y=0$	$\frac{35}{101}$	$\frac{20}{101}$	$\frac{55}{101}$
$Y=1$	$\frac{25}{101}$	$\frac{21}{101}$	$\frac{46}{101}$
Total	$\frac{60}{101}$	$\frac{41}{101}$	$\frac{101}{101}$

For the condition distribution of $X|Y=1$:

$$P(X=0|Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{\frac{25}{101}}{\frac{46}{101}} = \frac{25}{46}$$

$$P(X=1|Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{\frac{21}{101}}{\frac{46}{101}} = \frac{21}{46}$$

Activity: Weighted Coin Flip and Dice Roll

Suppose you flip a weighted coin (1 for heads, 0 for tails) and roll a dice simultaneously as an experiment over 200 times. We get the following results:

	$X=1$	$X=2$	$X=3$	$X=4$	$X=5$	$X=6$	Total
$Y=0$	$\frac{11}{200}$	$\frac{14}{200}$	$\frac{10}{200}$	$\frac{19}{200}$	$\frac{16}{200}$	$\frac{15}{200}$	
$Y=1$	$\frac{17}{200}$	$\frac{16}{200}$	$\frac{13}{200}$	$\frac{29}{200}$	$\frac{17}{200}$	$\frac{23}{200}$	
Total							

- Find $f_X(X)$ and $f_Y(y)$.
- Find $P(X=4|Y=1)$ and $P(X=4|Y=0)$.
- If X and Y were independent, what should we expect to see in our joint distribution?

Discrete to Continuous

As we saw with singular random variables when we went from discrete to continuous, you will see the switch from summation to integration here.

Additionally, we will be working with double integrals to find the **volume** under the surface. Hence, we will see this for some $A \in \mathbb{R}^2$:

$$P((X, Y) \in A) = \int_A \int f_{X,Y}(x, y)\, dx\, dy$$

Continuous Joint RVs - Joint CDF

For a single continuous random variable X, we had the following CDF:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$$

Definition (Joint CDF)

For two continuous random variables X and Y, the joint CDF is:

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f_{X,Y}(u,v)\,du\,dv$$

Continuous Joint RVs - Joint PDF

For a single continuous random variable X, we had the following PDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Definition (7.1.13: Joint PDF)

For two continuous random variables X and Y, the joint PDF is the derivative of the joint CDF with respect to x and y:

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

Continuous Joint RVs - Marginal PDF

If we want to find the marginal PDF of X, we will integrate the density under a curve at some $X = x$. As such, we only need to take one integral.

Definition (7.1.14: Marginal PDF)

For two continuous random variables X and Y with joint PDF $f_{X,Y}$, the marginal PDF of X is:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$$

Continuous Joint RVs - Conditional PDF

From a joint distribution, we may care about point probabilities of one random variable given the value of a different random variable. We call this single distribution a **conditional** distribution.

Definition (7.1.15: Conditional PDF)

For continuous RVs X and Y with joint PDF $f_{X,Y}$, the conditional PDF of Y given $X = x$ is:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Example: Uniform Density on a Plane

Suppose (X, Y) is a random point uniformly distributed over the triangle defined by:

$$\{(x, y) : 0 \leq y < x \leq 1\}.$$

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To determine the joint PDF, we use the fact that the total volume under the density function must equal 1.

- The area of the triangular region is $\frac{1}{2}$.
- Therefore, the height must be 2 so that the total volume equals 1.

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- Therefore, the height must be 2 so that the total volume equals 1.

Thus, our joint PDF is

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 \leq y < x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example: Uniform Density on a Plane

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \leq y < x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

To find the marginal PDF of X, we can integrate:

$$\begin{aligned} f_X(x) &= \int_0^x 2dy \\ &= 2y\Big|_0^x \\ &= \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Example: Uniform Density on a Plane

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \le y < x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

To find the conditional PDF of $Y|X = x$, we can use the definition:

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f_{X,Y}(x,y)}{f_X(x)} \\ &= \frac{2}{2x} \\ &= \begin{cases} \frac{1}{x}, & 0 \le y < 1, 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Activity: Uniform Density on a Plane

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \leq y < x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the marginal PDF of Y.
2. Find the conditional PDF of $X|Y = y$.
3. Sketch out the joint PDF of X, Y.

Example: Uniform Density on a Plane

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \le y < x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

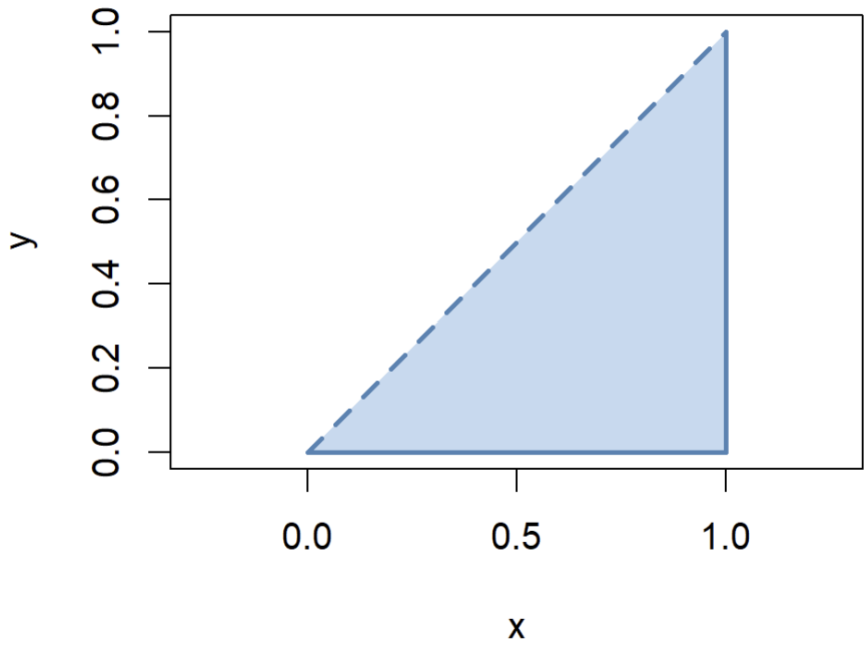


Figure: Bird's Eye View of Joint PDF

Hybrid Situations

Look through Example 7.1.26 in the *Introduction* to Probability textbook for more details.

- For the joint probability of a continuous RV X and discrete RV Y: an expression in terms of X
- For the marginal probabilities: use of Law of Total Probability to find the marginal CDF:

$$F_X(x) = P(X \leq x) = \sum_y P(X \leq x | Y = y) P(Y = y)$$

- Combination of both summation and integration

Questions to Ponder

- How would the definitions I have shown change if we were working with $3, 4, 5, \ldots, N$ random variables jointly?
- How might one find an expectation of two jointly random variables?

Next Time

- What makes independent joint random variables?
- Bayes rule and Law of Total Probability with joint random variables
- 2D LOTUS
- Covariance and correlation